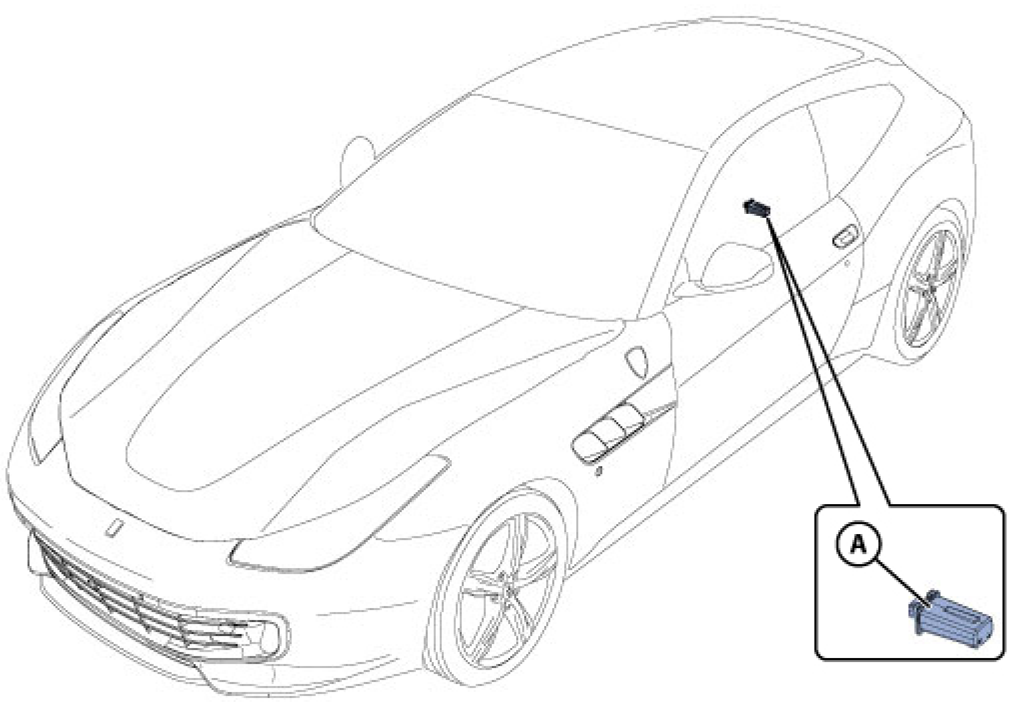
GTC4Lusso USA - TPMS tyre pressure and temperature monitoring system - Latest Update: 2021-01-26

D2.03 TPMS tyre pressure and temperature monitoring system

TPMS system layout



A - Antenna

TPMS system components

Sensors

Fastened to the rims, inside the tyres.

The sensors measure the temperature and pressure within the tyre.
The sensors transmit data to the digital antenna as an RF signal (433.92 MHz / 315 MHz).

Digital antenna / ECU

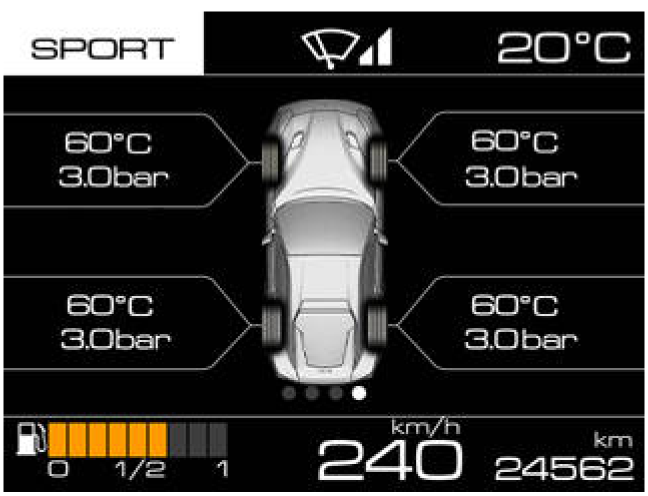
Fastened to the floorpan of the vehicle, at the rear.
Receives data from the sensors, and decodes and processes the data to provide the instrument panel with the information needed.

System function

For increased safety, the vehicle is equipped with a tyre pressure monitoring system (TPMS), which warns the driver by illuminating the relative warning lamp in the event of low pressure in one or more tyres.
When a low tyre pressure condition is indicated by the TPMS warning lamp, stop the vehicle as soon as possible, check the tyres and inflate them to the correct pressure.
Driving with an insufficiently inflated tyre may cause the tyre to overheat and will eventually result in tyre damage.
Insufficient tyre pressure also increases fuel consumption, reduces tyre durability and may also compromise the braking performance and handling of the vehicle.
The system measures the pressure and temperature of the tyres with sensors fixed to the interior of the wheel rims, next to the inflation valve.
These sensors transmit a signal received by the antenna, which also functions as an ECU.
The ECU processes the signal received from the sensors and transmits a set of information to the instrument panel regarding the pressure and temperature status of the tyres and any system errors.

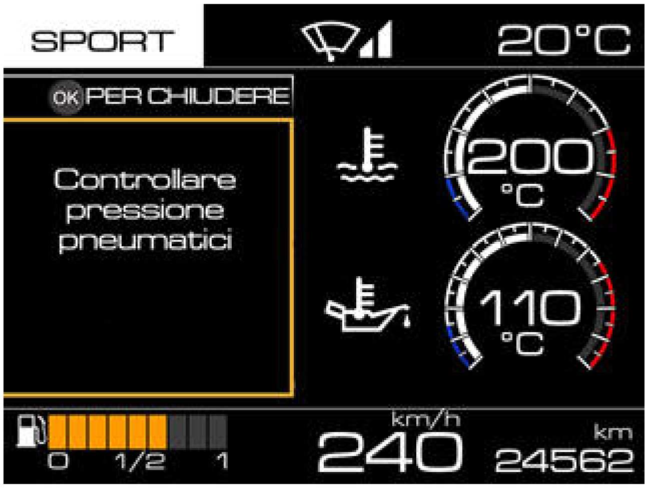
Information on TFT display

The system may be subject to temporary radio frequency disturbance caused by devices using similar frequencies.

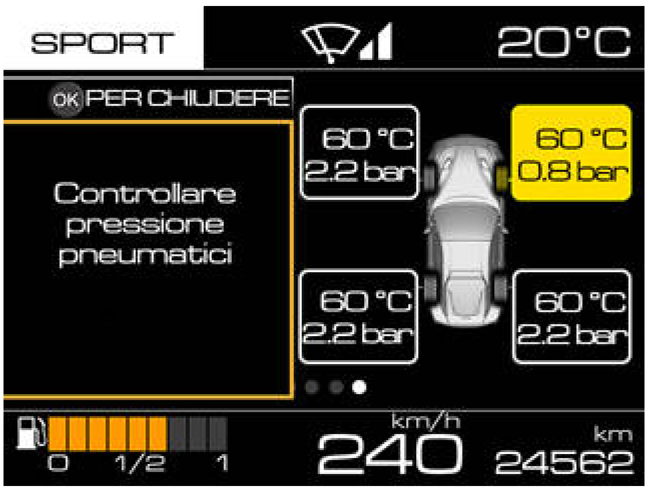


TYRES screen: The relative controls on the left hand TFT display may be used by the driver to access the informative screen displaying a vehicle symbol indicating the pressure and temperature values for each tyre.

- Insufficient pressure of one or more tyres

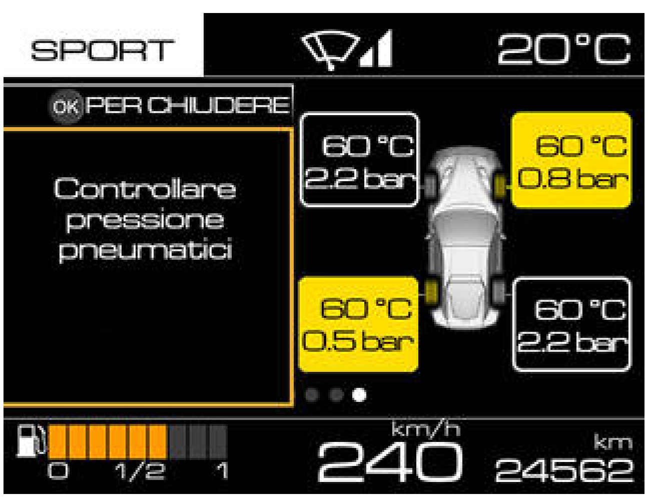


With any currently active screen, when the instrument panel receives the signal from the electronic control unit indicating that the pressure of one or more tyres is below the warning threshold, but the system cannot recognise which wheel is generating the failure signal, the TFT display displays the specific message "**Check tyre pressure**" in area B2 of the screen, without the vehicle symbol.



With any currently active screen, when the instrument panel receives the signal from the electronic control unit indicating that the pressure of one or more tyres is below the warning threshold, the TFT display displays the specific message "**Check tyre pressure**" in area B2 of the screen together with the vehicle symbol in area B1 indicating tyre pressures.

The background of the box indicating the pressure and relative unit of measurement for the faulty tyre turns amber.



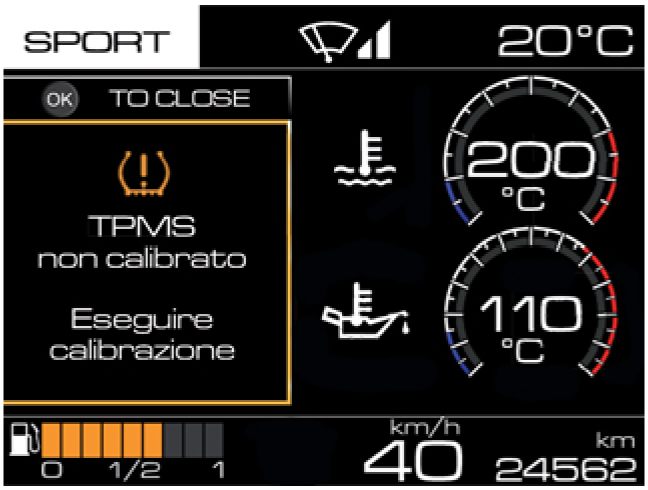
With any currently active screen, when the instrument panel receives the signal from the electronic control unit indicating that the pressure of more than one tyre is below the warning threshold, the TFT display displays the specific message “**Check tyre pressure**” in area B2 of the screen together with the vehicle symbol in area B1 indicating tyre pressures.

The background of the box indicating the pressure and relative unit of measurement for the faulty tyre turns amber.

The display cycle is 20 seconds, after which time the specific screen disappears.

- TPMS not calibrated

The system must be calibrated from the relative menu option on the left hand TFT display after replacing or inflating any of the tyres.



When the TPMS system has not been calibrated or a tyre has been replaced, the TFT display displays the specific message "**TPMS not calibrated**" together with the message "**Execute calibration**".

The specific warning lamp starts flashing simultaneously, and continues to flash for 90 seconds.

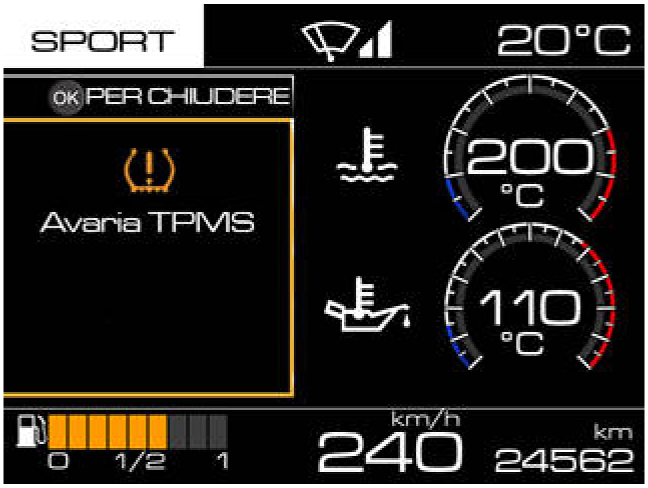
The symbol and the message are cleared at the end of the 20 second display cycle and the previously active screen is displayed again, while the indicator remains lit until calibration is complete.

The TPMS system may be calibrated from the specific Menu option on the left hand TFT display, with the instrument panel on and the engine off.

To calibrate the TPMS system, recall the MENU screen of the left hand TFT display with the instrument panel on and the engine off. From the MENU screen, select the options “**Car setup**” and “**Calibrate TPMS**”.

Once the menu option has been selected and the request for the calibrated procedure is subsequently accepted by the system, the TFT display displays the specific message "**Calibration activated**" for 5 seconds.

- TPMS failure

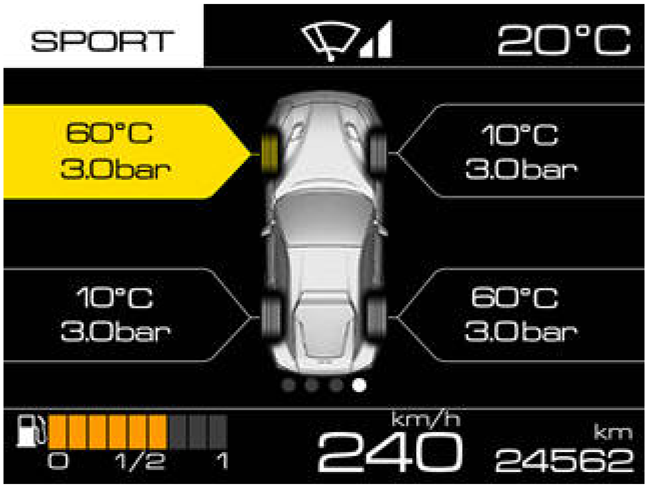


In the event of a fault in the circuit and/or wiring harness leading to the ECU, or in the event of a TPMS ECU fault, or in the event of no signal reception from one or more sensors due to fault, breakage or flat battery, the TFT display displays the specific message "**TPMS failure**".

The specific warning lamp starts flashing simultaneously, and continues to flash for 90 seconds.
The TYRES screen is not recallable by the driver.

- Punctured tyre

When the instrument panel receives the signal from the electronic control unit indicating that the pressure of one or more normal tyres is below the alarm threshold, but the system cannot recognise which wheel is generating the failure signal, the TFT display displays the specific message “**Low tyre pressure**” together with the message "**Do not proceed**" in area B2 of the screen, without the vehicle symbol.
Simultaneously, the specific warning light is displayed continuously on the dial.
The display cycle for the specific screen is 20 seconds. After this period, the previously active screen is shown on the display, while the indicator on the dial remains continuously lit.



When the instrument panel receives the signal from the electronic control unit indicating that the pressure of one or more normal tyres is below the alarm threshold, the TFT display displays the specific message “**Low tyre pressure**” together with the message "**Do not proceed**" in area B2 of the screen, with the vehicle symbol in area B1 indicating tyre pressures.

Simultaneously, the specific warning light is displayed continuously on the dial.

The display cycle for the screen is 20 seconds. After this period, the previously active screen is shown on the display, while the indicator on the dial remains continuously lit.

The background of the box indicating the pressure and relative unit of measurement for the faulty tyre turns amber.

- TPMS temporarily inactive

The screen shown is displayed in the following cases:

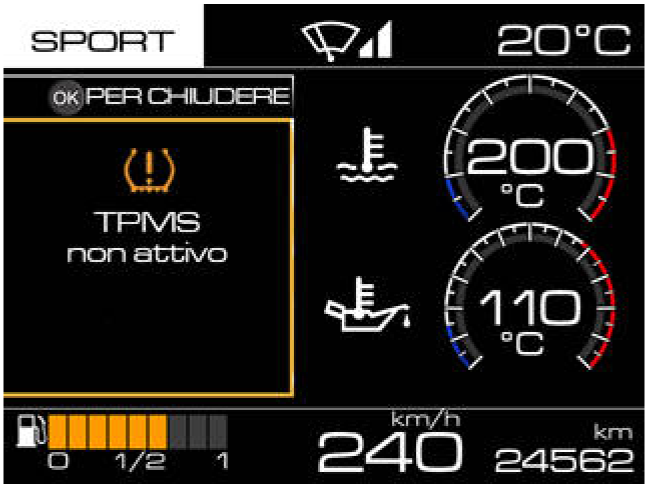
- sensor overheat;
- during calibration (TPMS ECU does not recognise the sensors);
- radio frequency interference of wheel sensor signals.

Simultaneously, the relative indicator on the dial starts flashing, and continues to flash for 90 seconds.

After this period, the indicator remains continuously lit until normal operating conditions are restored.

The TYRES screen is not recallable by the driver.

- "TPMS inactive"



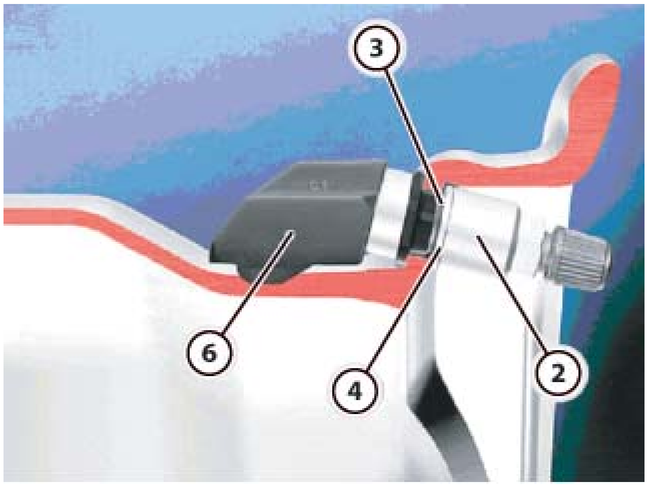
When the system has been disabled via the diagnostic tester, the TFT display displays the specific message "**TPMS inactive**" when the instrument panel is switched on.
Simultaneously, the relative indicator on the dial starts flashing, and continues to flash for 90 seconds.
After this period, the indicator remains continuously lit until normal operating conditions are restored.
The TYRES screen is not recallable by the driver.

Removing the tyre pressure sensors

ALWAYS replace with new components when REMOVING the TPMS sensor, complete with valve, from the rim.
The sealant liquid contained in puncture repair aerosols may damage the sensor. Always replace the sensor if a tyre has been repaired with a puncture repair aerosol.
Replace the entire sensor-valve assembly even if only one of the components is damaged.

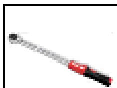
➤ Remove the tyre (D2.02).

- To prevent the inflation valve from turning, insert the rod **(1)**, included in the kit, in the hole on the sensor side.



- Undo the nut **(2)** from the inflation valve, retrieving the insulator washer **(3)**.
- Remove the sensor **(6)** with valve **(4)** from the wheel rim and replace.

Refitting the tyre pressure sensors



Tightening torque		Nm	Class
Fastening tyre inflation valve	Nut	4 Nm	B
Fastening tyre pressure sensor	Screw	4 Nm	B

When replacing the tyre pressure sensors, carry out the new ECU sensor acquisition procedure with the DEIS tester.

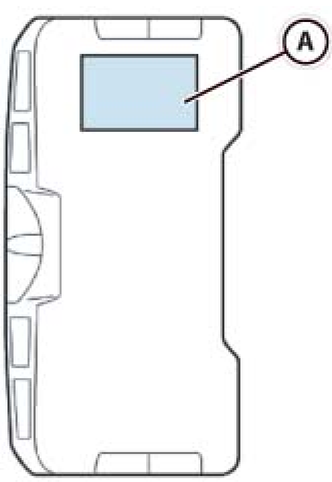
Always finish all the components in the batch opened previously before starting a new sensor batch, as the sensors are equipped with a battery with a limited lifespan.

Once the sensor has been activated, the internal battery has a lifespan of approximately 10 years.

ALWAYS replace with new components when REMOVING the TPMS sensor, complete with valve, from the rim.

The sealant liquid contained in puncture repair aerosols may damage the sensor. Always replace the sensor if a tyre has been repaired with a puncture repair aerosol.

Replace the entire sensor-valve assembly even if only one of the components is damaged.



The sensor must not be treated with detergents, solvents or other harsh products.

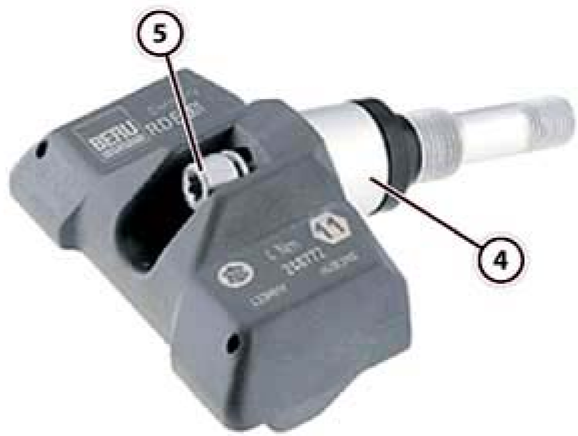
Do not clean with compressed air or pressurised liquids (e.g. pulivapor).

Take particular care not to tear the surface of the filter interface (A) on the inner side of the sensor. Always replace the sensor if the interface surface is dirty or torn.

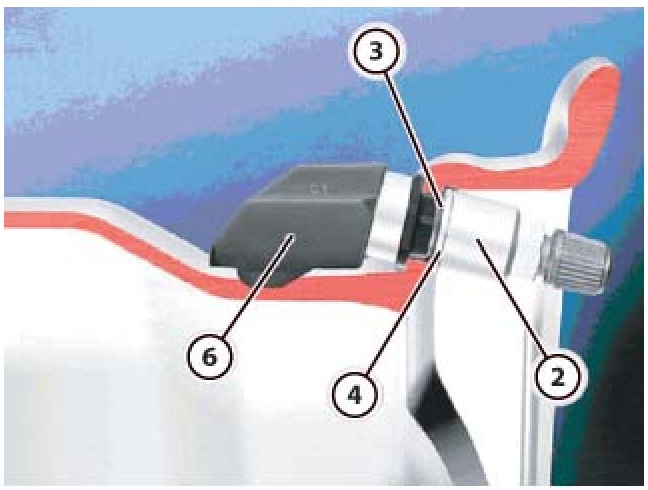
Store all the components of the sensor-valve assembly in a clean, dry place.

- Select a new TPMS valve-sensor assembly.

- Clean the sensor mating area on the wheel rim.
- Check that the sensor frequency matches the system frequency: **433 MHz** for the **STANDARD** version, white label on sensor; **315 MHz** for the **JAPAN** version, blue label on sensor.



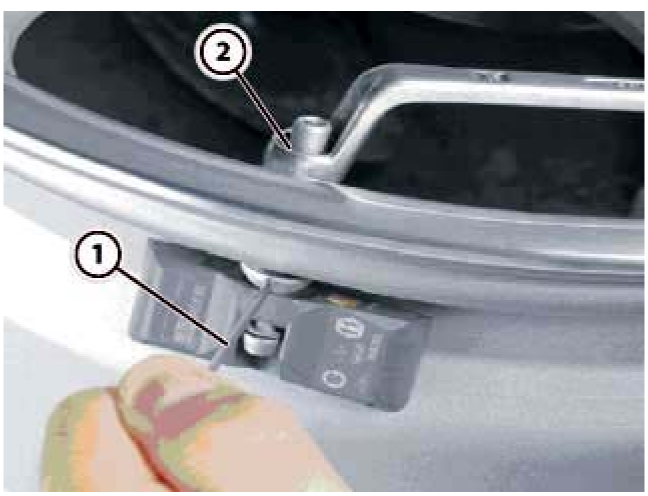
- Fit the self-locking screw **(5)** in the hole on the sensor and tighten the valve **(4)** by two or three turns.



- Fit the valve **(4)** with sensor **(6)** already assembled, in the hole on the wheel rim.

Ensure that the insulating washer is fitted onto the outside of the rim and is perpendicular to the valve.

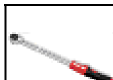
- Fit the insulating washer **(3)** and tighten the nut **(2)** fully.



Use a rod to ensure that the hole on the valve is kept outside the rim.

- Fit the rod **(1)** in the radial hole of the valve and tighten the nut **(2)**.

ALWAYS use a torque wrench, using only certified and calibrated tools to ensure that the correct tightening torque is achieved.



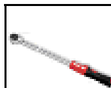
Tightening torque	Nm	Class
Nut	4 Nm	B

- Remove the rod **(1)**.

For the sensor to function correctly, there must be contact between the rim and the mounting points of the sensor itself. The system may not function correctly if the sensor is not installed exactly as instructed.

- Push the sensor gently onto the rim to ensure contact between the rim and the mating surface of the sensor itself.
- Tighten the self-locking screw **(5)**.

ALWAYS use a torque wrench, using only certified and calibrated tools to ensure that the correct tightening torque is achieved.



Tightening torque	Nm	Class
Screw	4 Nm	B

➤ Refit the tyre (D2.02).

Activating the tyre pressure sensor

Always finish all the components in the batch opened previously before starting a new sensor batch, as the sensors are equipped with a battery with a limited lifespan.
Once the sensor has been activated, the internal battery has a lifespan of approximately 10 years.

During manufacture, the tyre pressure sensor is put into a dormant state to preserve battery life during storage.

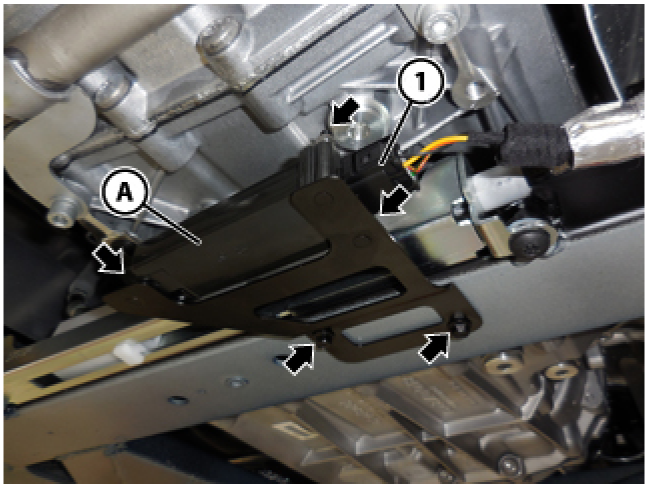
The sensor is activated when it detects a given pressure. To activate, proceed as follows.

- Refit the tyre pressure sensors (D2.03).
- Refit the tyre (D2.02).
- Inflate the tyre to a pressure greater than **0.6 bar (8.7 psi)** and keep inflated at this pressure **for a few minutes**. This causes the sensor to activate.
- Inflate the tyre to the specified pressure.
- Refit the complete wheels (D2.01).

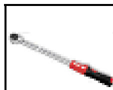
Replacing the TPMS system antenna

Tightening torque		Nm	Class
Fastening the TPMS antenna	Nut	1.25 Nm	C
	Screw	5 Nm	B

- Disconnect the battery (F2.01).
- Remove the flat undertray sections (E3.13).
  - Remove the rear flat undertray section.

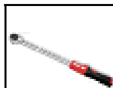


- Disconnect the connector **(1)**.
- Undo the indicated screws and remove the bracket from the chassis.
- Undo the indicated nuts, remove the antenna **(A)** and replace.
- Fit the new antenna **(A)** on the bracket and tighten the indicated nuts.



Tightening torque	Nm	Class
Nut	1.25 Nm	C

- Fit the bracket on the chassis and tighten the indicated screws.



Tightening torque	Nm	Class
Screw	5 Nm	B

- Connect the connector **(1)**.

- Refit the flat undertray sections (E3.13).
  - Refit the rear flat undertray section.

- Connect the battery (F2.01).